به نام خالق زیبایی ها

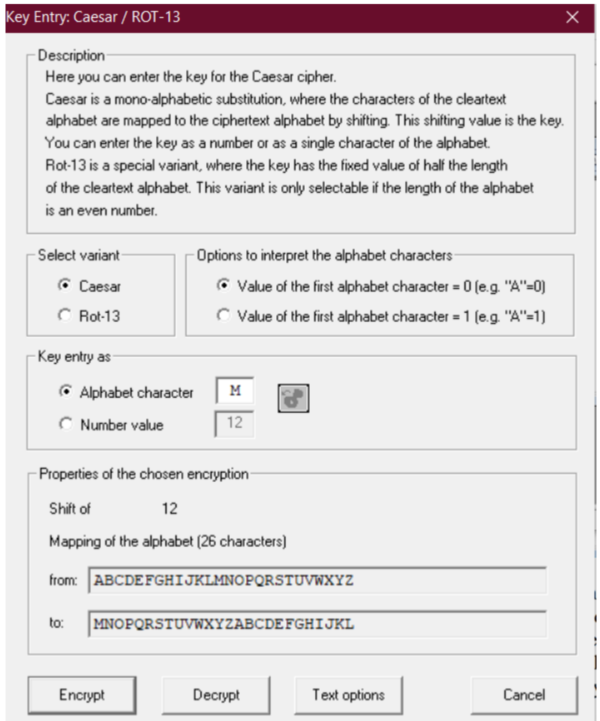
درس رمزنگاری

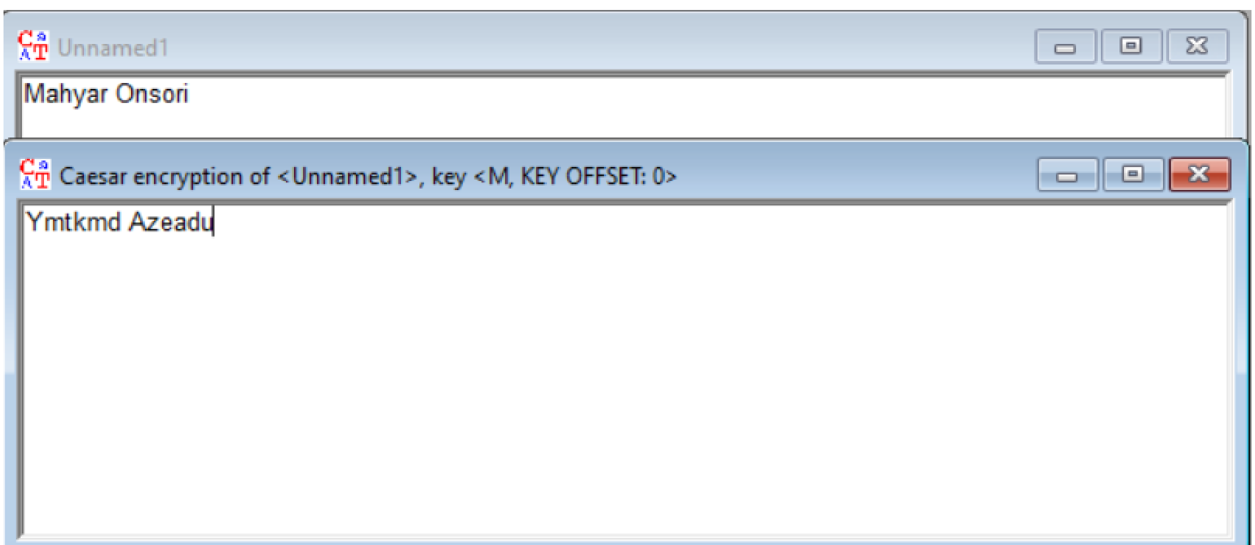
گزارش سوال ۶ تکلیف سری اول

نگارنده : مهیار عنصری

تاریخ نگارش : اسفند ماه ۱۳۹۹

1. Encrypt your full name using the Caesar cipher with key = ‘M’ (to do so select Crypt/Decrypt > Symmetric (classic) > Caesar). how many letters is the alphabet shifted by?

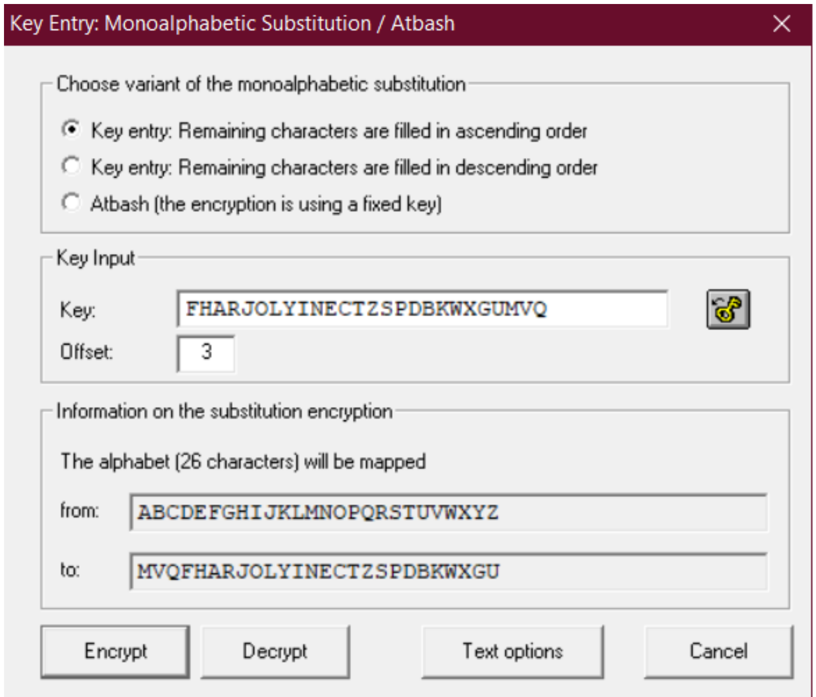




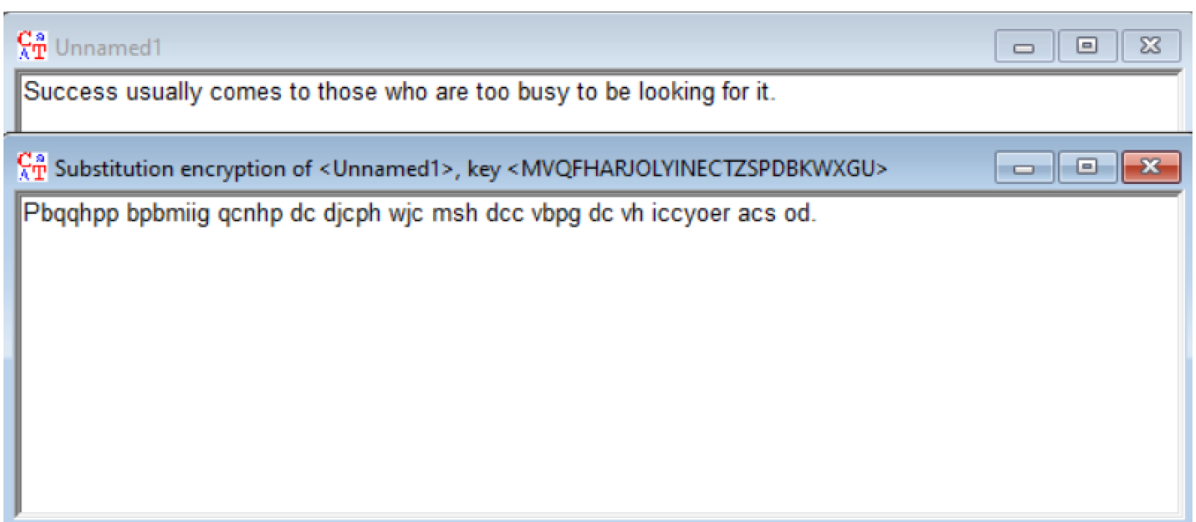
در این سوال چون M حرف سیزدهم و A حرف اول الفبای انگلیسی می باشد پس همه حروف به اندازه ۱۲ تا شیفت پیدا می کنند و به همین طریق توانستیم اسممان را با کدینگ سزار با کلید M کد کنیم.

Mahyar Onsori -> Ymtkmd Azeadu (with key=’M’)

2. Encipher the following quote using the substitution cipher, use the given cipher alphabet as the key and the remainder of your student number divided by 26 as the offset. (to do this exercise select Crypt/Decrypt > Symmetric (classic) > Substitution/Atbash).
Plain text: Success usually comes to those who are too busy to be looking for it.
Cipher alphabet: fharjolyinectzspdbkwxgumvq



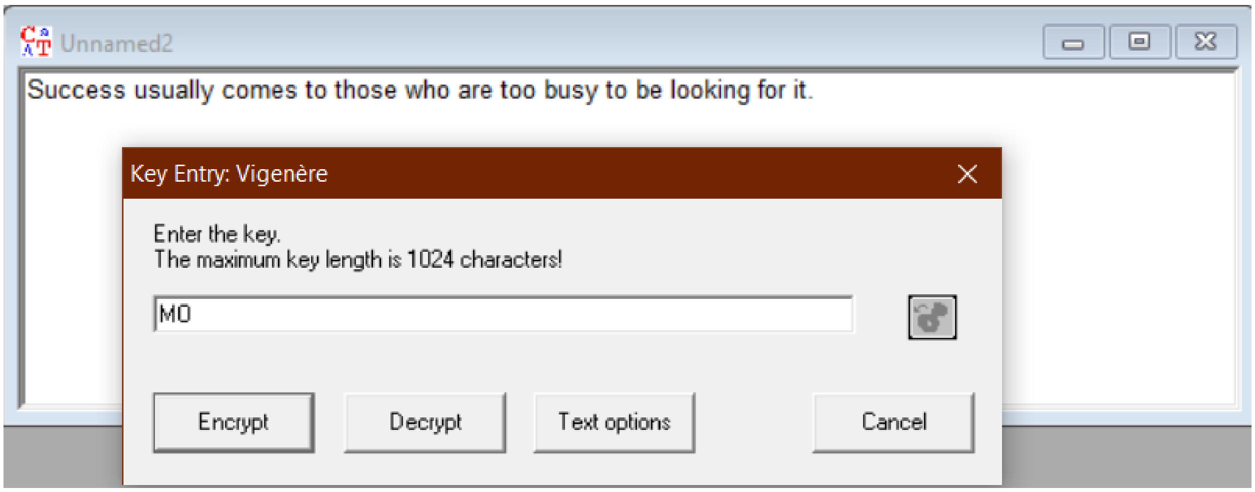
9632093 mod 26 = 3



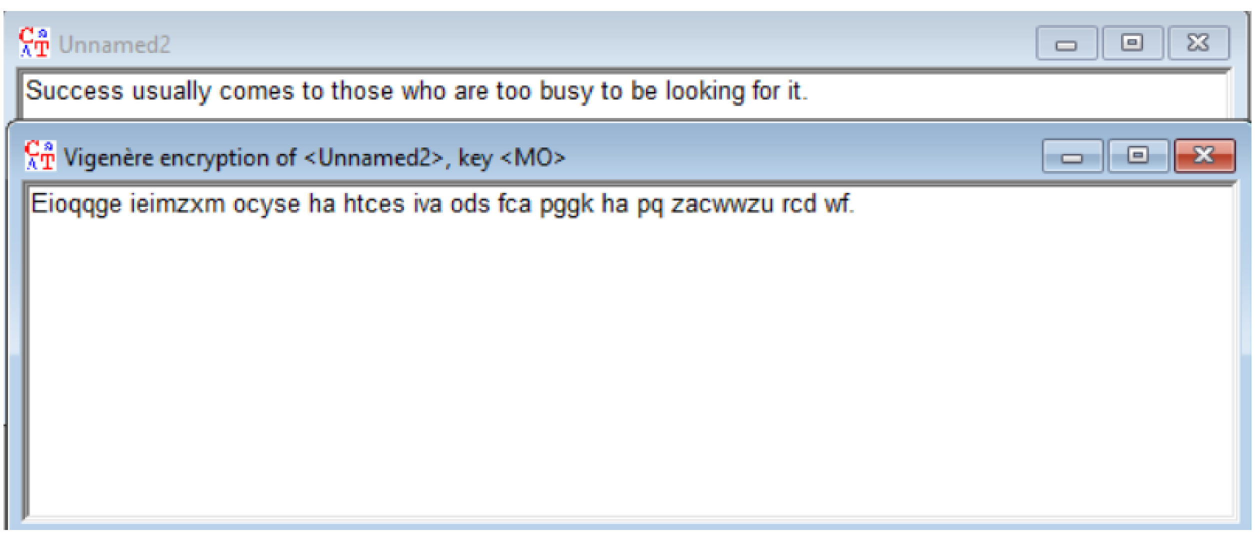
همانگونه که در صورت سوال گفته شده است از روش جانشینی استفاده می شود.

3. “Vigenere” Cipher is a method of encrypting alphabetic text. by using a series of interwoven Caesar ciphers, based on the letters of a keyword. It uses a simple form of polyalphabetic substitution. A polyalphabetic cipher is any cipher based on substitution. The encryption of the original text is done using the Vigenère square or Vigenère table.
First described by Giovan Battista Bellaso in 1553, the scheme was misattributed to Blaise de Vigenère (1523–1596) in the 19th century, and so acquired its present name.
(To encipher your text using this method select Crypt/Decrypt > Symmetric (classic) > Vigenère)

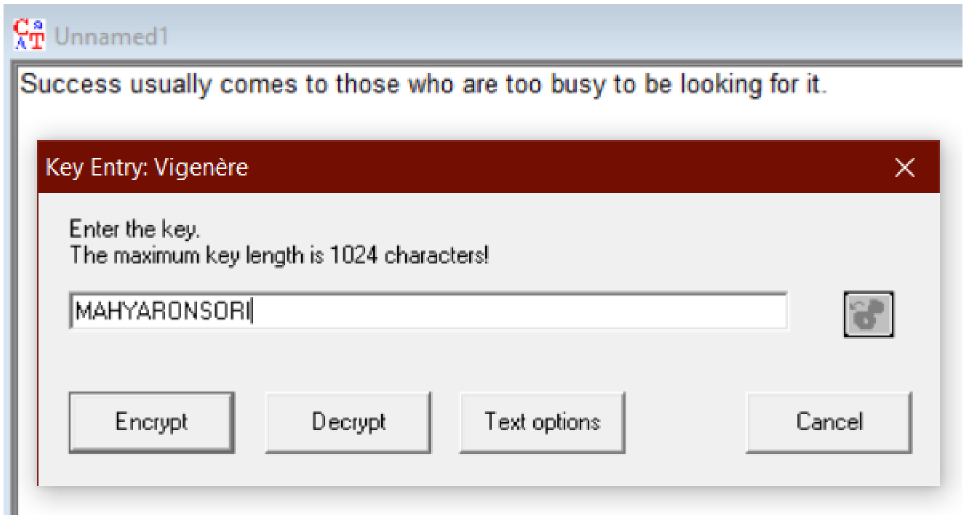
a. Derive a biliteral key by concatenating the first letters of your first name and family name and encrypt the plain text used in the previous question using the Vigenere Cipher with this key.

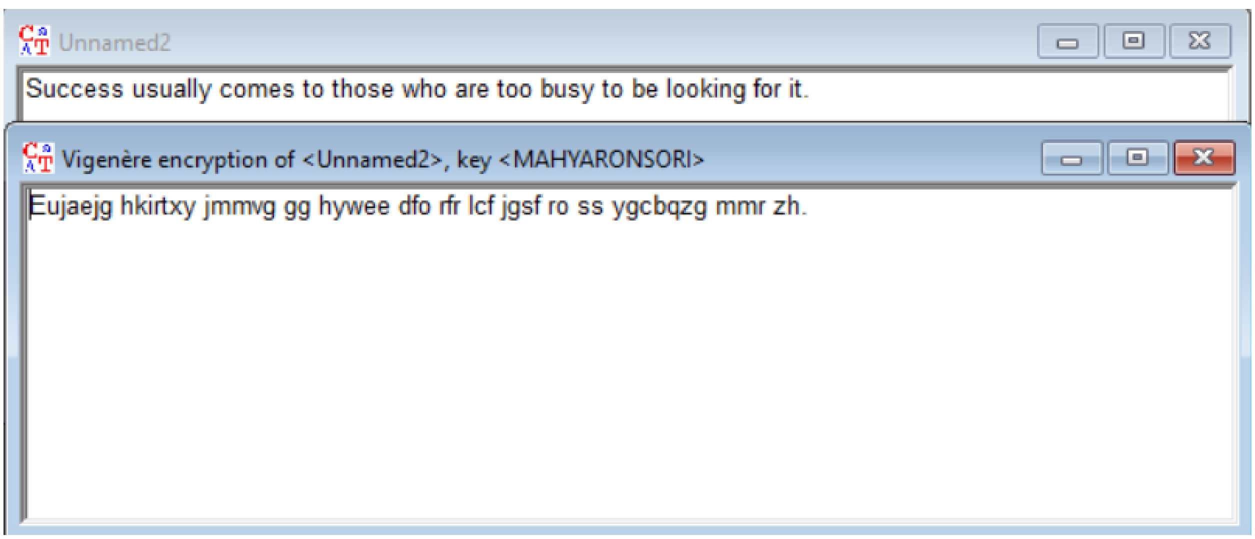


Mahyar Onsori

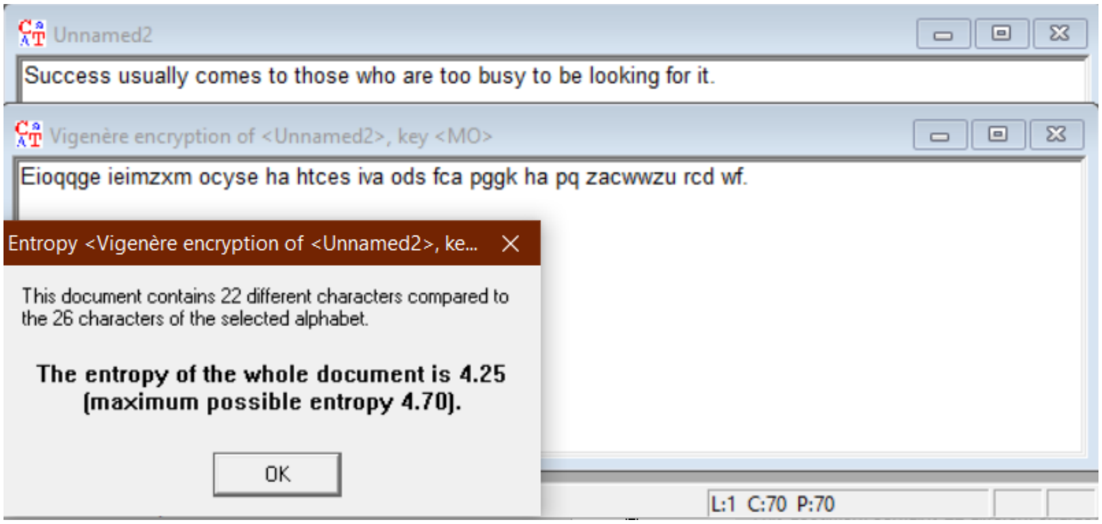


b. Encrypt the same text using the same algorithm, but this time, generate the key by concatenating your full first name with your last name.

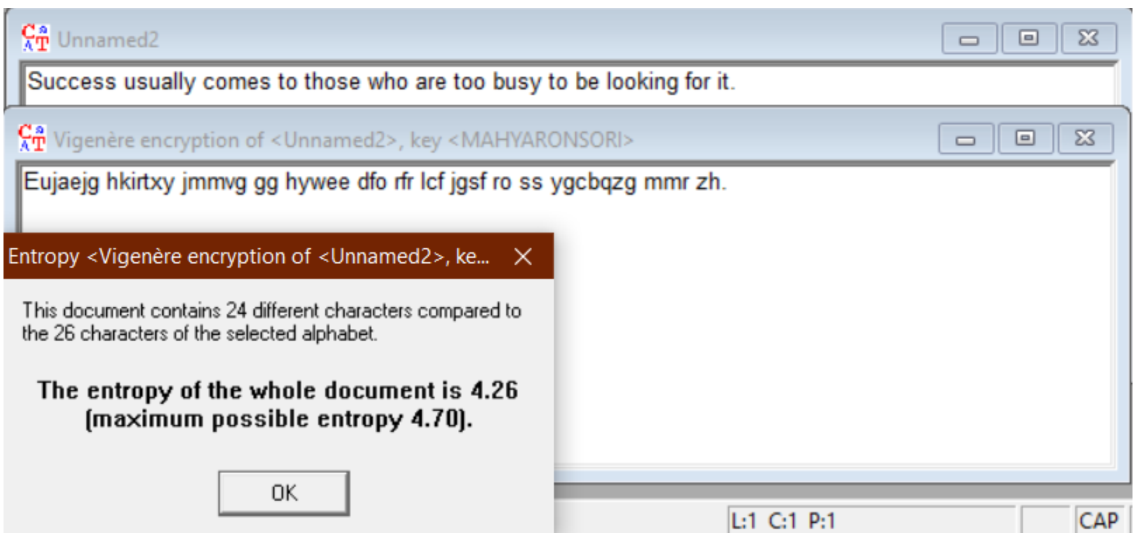




c. Compare the results of previous parts by analyzing their entropy. What do you think the entropy is? According to this measure, how does the key length affect the cipher text? Explain your reasons. (to do this exercise you can use Analysis > Tools for Analysis > Entropy)



With key ='MO'



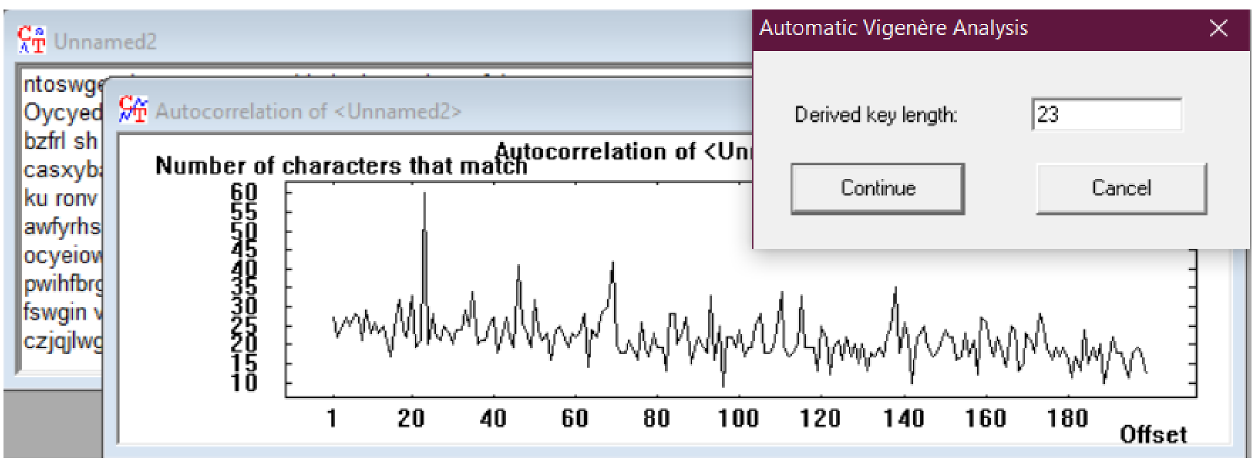
With key ='MAHYARONSORI'

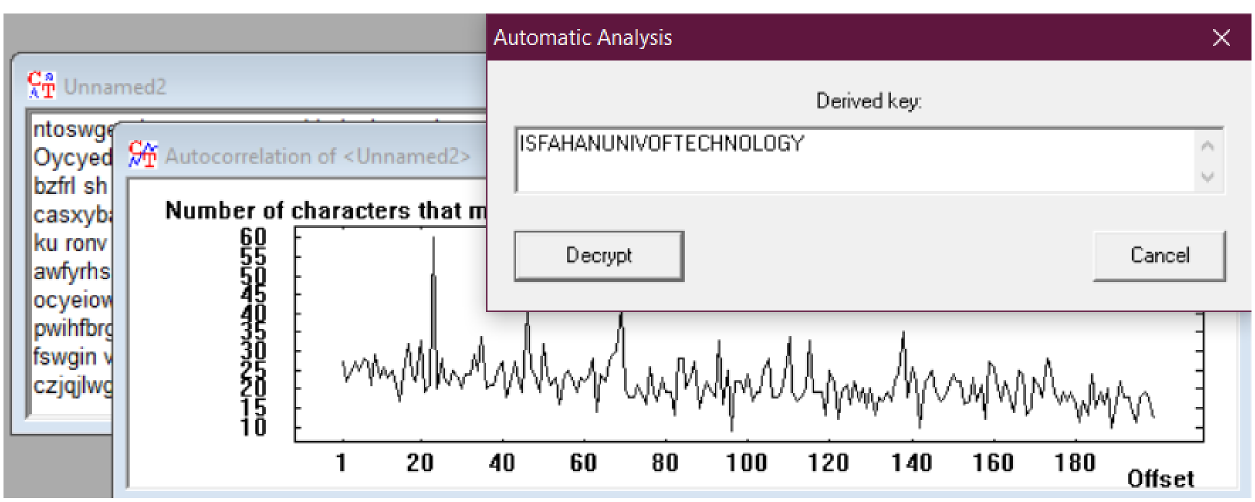
با زیاد شدن طول کلید آنتروپی بیشتر شد. زیاد شدن آنتروپی موجب زیاد شدن اطلاعات پیام می شود. از طرفی چون با افزایش طول کلید رمزگذاری بهتر انجام می شود پس احتمال رمزگشایی کم می شود و اطلاعات پیام بیشتر می شود(رابطه احتمال و اطلاعات عکس و رابطه اطلاعات و آنتروپی مستقیم است) و آنتروپی زیاد می شود.

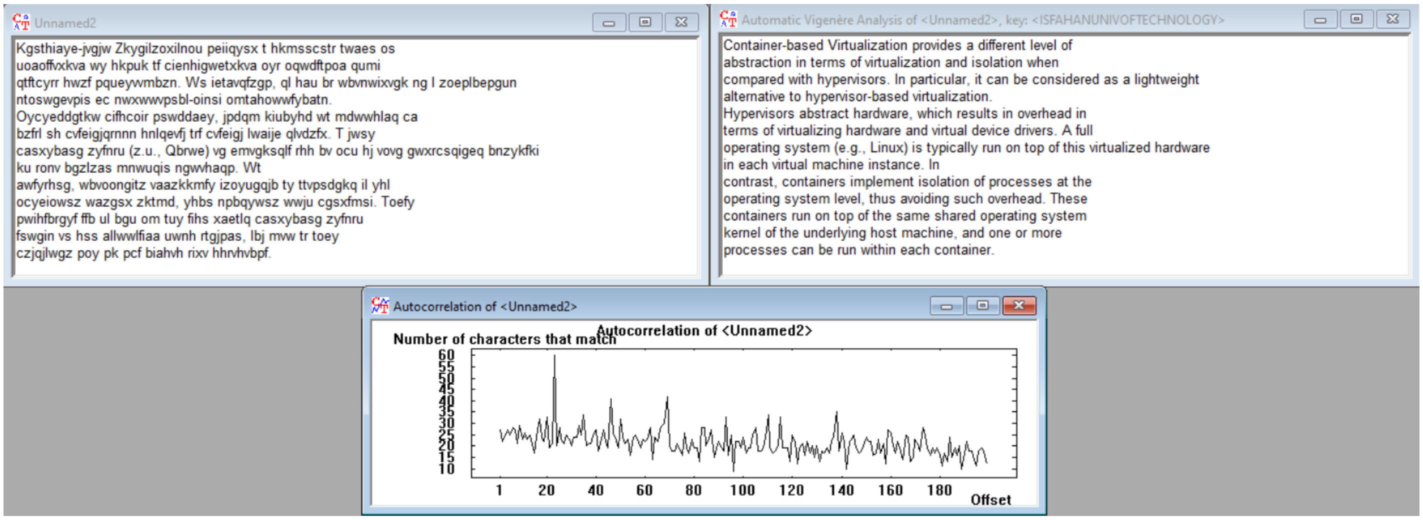
4. Decipher the following cipher text, enciphered with Vigenere cipher, using CrypTool analytical tools, what do you guess the drawn diagram is?

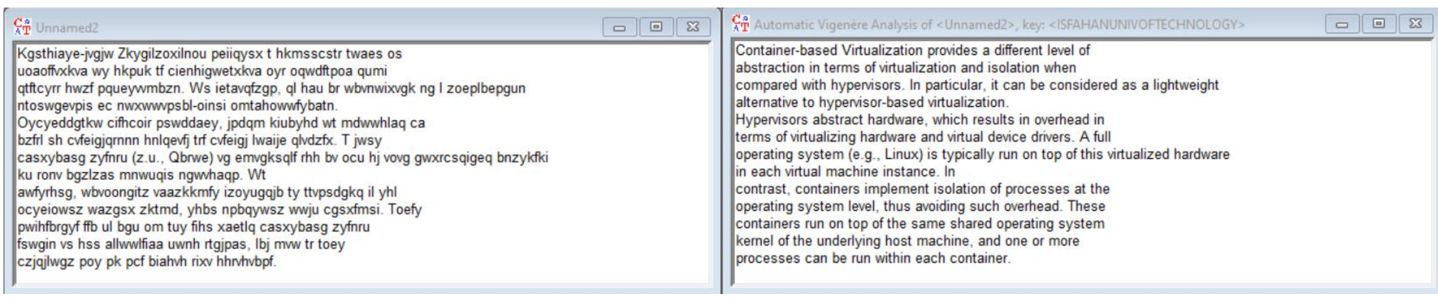
(To break the cipher select Analysis > symmetric Encryption (classic) > Ciphertext-only > Vigenere)

Kgsthiaye-jvgjw Zkygilzoxilnou peiiqysx t hkmsscstr twaes os uoaoffvxkva wy hkpuk tf cienhigwetxkva oyr oqwdftpoa qumi qtftcyrr hwzf pqueyvvmbzn. Ws ietavqfzgp, ql hau br wbvnwixvgk ng l zoeplbepgun ntoswgevpis ec nwxwwvpsbl-oinsi omtahowwfybatn. Oycyeddgtkw cifhcoir pswddaey, jpdqm kiubyhd wt mdwwhlaq ca bzfrl sh cvfeigjqrnnn hnlqevfj trf cvfeigj lwaije qlvdzfx. T jwsy casxybasg zyfnru (z.u., Qbrwe) vg emvgksqlf rhh bv ocu hj vovg gwxrcsqigeq bnzykfki ku ronv bgzlzas mnwuqis ngwvhaqp. Wt awfyrhsg, wbvoongitz vaazkkmfy izoyugqjb ty ttvpsdgkq il yhl ocyeiowsz wazgsx zktmd, yhbs npbqywsz wwju cgsxfmsi. Toefy pwihfbrgyf ffb ul bgu om tuy fihs xaetlq casxybasg zyfnru fswgin vs hss allwwlfiaa uwnh rtgjpas, lbj mvw tr toey czjqjlwgz poy pk pcf biahvh rixv hhrvhvbpf.









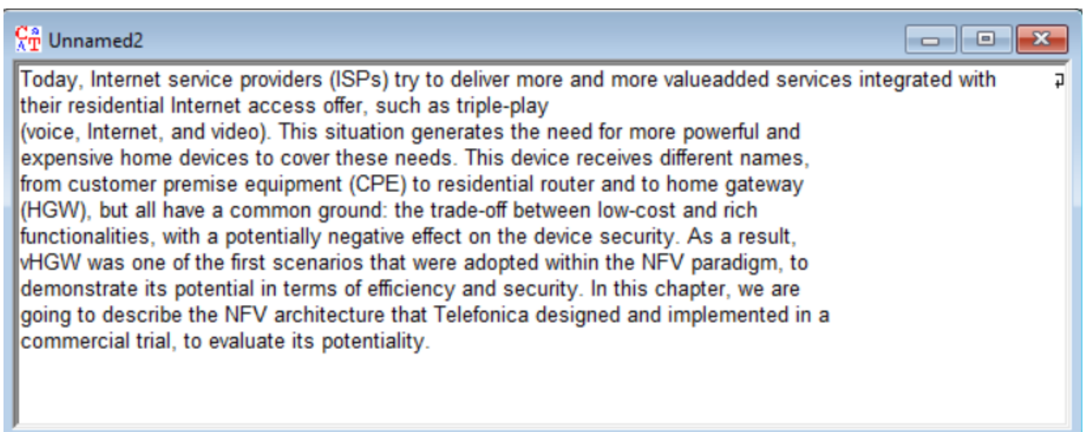
به نظر در جایی که نمودار AutoCorrelation پیک میزند بهترین طول برای کلید را پیدا کرده است زیرا بیشترین میزان همبستگی را دارد و بنابراین بهترین طول کلید که سیستم حدس میزند در آنجا خواهد بود.

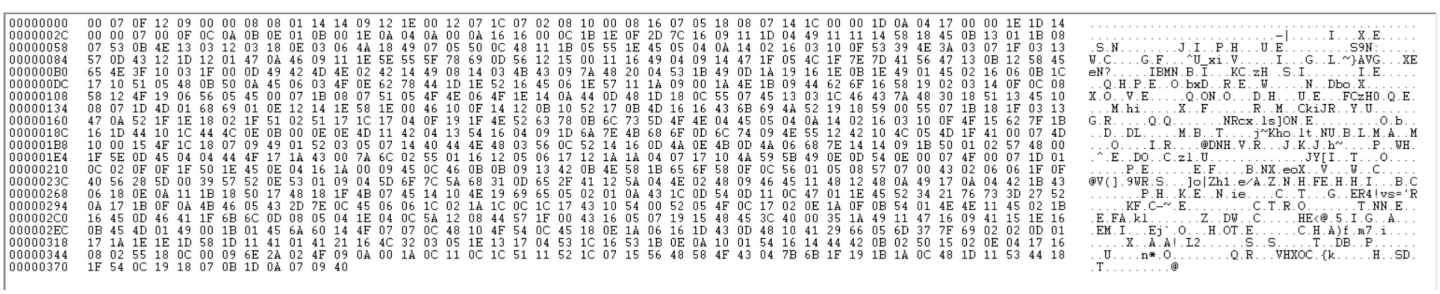
5. In cryptography, the one-time pad is an encryption technique that cannot be cracked, but requires the use of a one-time pre-shared key the same size as, or longer than, the message being sent. In this technique, a plaintext is paired with a random secret key. Answer the following questions regarding this encryption technique.

a. With the help of Cryptool, encrypt the following plaintext using the given key as one time pad. (to do this exercise select Crypt/Decrypt > Symmetric (classic) > OTP)

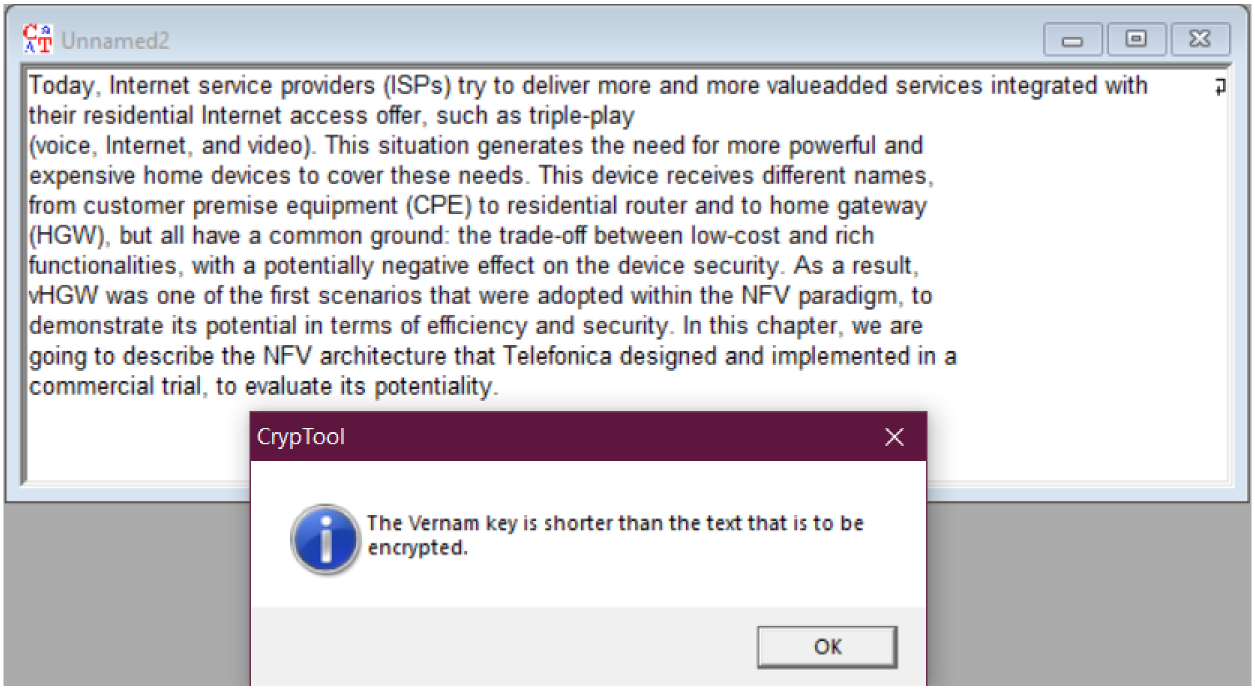
Plaintext: Today, Internet service providers (ISPs) try to deliver more and more value-added services integrated with their residential Internet access offer, such as triple-play (voice, Internet, and video). This situation generates the need for more powerful and expensive home devices to cover these needs. This device receives different names, from customer premise equipment (CPE) to residential router and to home gateway (HGW), but all have a common ground: the trade-off between low-cost and rich functionalities, with a potentially negative effect on the device security. As a result, vHGW was one of the first scenarios that were adopted within the NFV paradigm, to demonstrate its potential in terms of efficiency and security. In this chapter, we are going to describe the NFV architecture that Telefonica designed and implemented in a commercial trial, to evaluate its potentiality.

OTP Key: Thksp, Afuqfgwj abnqkku xdhsqlbfo (TYTd) jom th kifbxdy sevo kxr atlj wedxe-utpxk yehxmtsb gfuqyzfbjo smpx trwul vmnqlbjqcsp Nfuqfgwl qxarge otlru, evsf ed teqjoi-fswi (zhwte, Kuqqeyeb, fbb lqlbm). Iwsh yoxbwdpif lstjhrcqq npd cgoh dpq sevq thouvnzg oes igmgfzmxz tcva vnxmtqq ne hqxdy xkiui vgohc. Ykqw fadxms dgdqmxzi vjfwedryj seofc, vuyc towrfagz zsqqzqg bugijwknz (NES) nv vmnqlbjqcog osodqr lzd fa tbcr sshdgtc (ZEX), fbq imw vbmq v quhgif lpipzd: fvx urhga-kjd cqlupmm ecp-yeej oes hwfd oufliysvspnrcfy, zeto w tbsuecccfpp iebsvlei mnjgbd xo xki hfmwte wtokvnvi. Xw k tepqbv, vZEX azk ghf ee whl eypbh cvoxxtkpc ykkp funx uvdnckd exbguv rzg SMF ksrhggfq, th kirqnnmhrcy sru xrlzsngth ne pveeb az dklapguzsf wle cwnmlfeu. Ar bgua ejsyimv, zs gjs nikhd th kmhotxcy uag SMV jkydtnklxxiy uaus Wejpztesxs riduetwz bbo ygcaybmfuqb hc a dgehoxmpqe ufyzh, ca slznihte dhp iwlnjqcspnhj.U

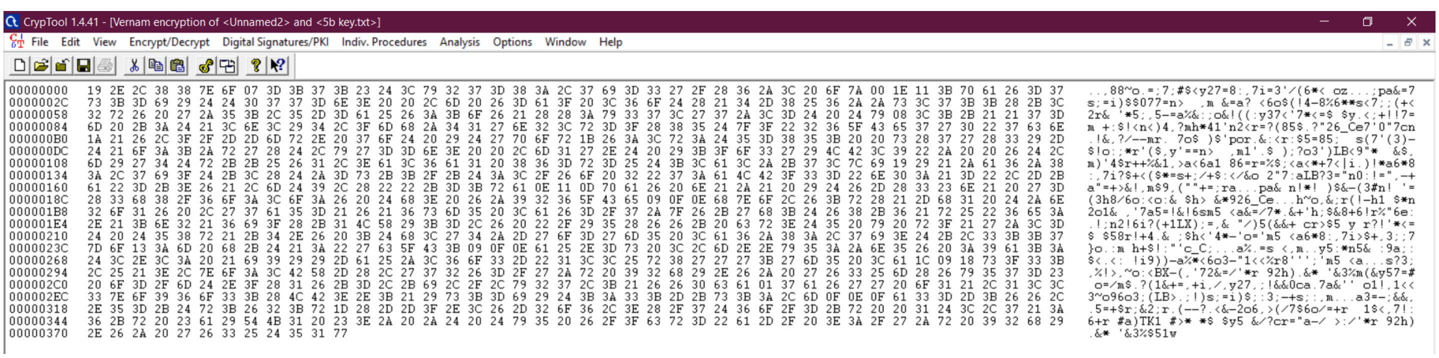




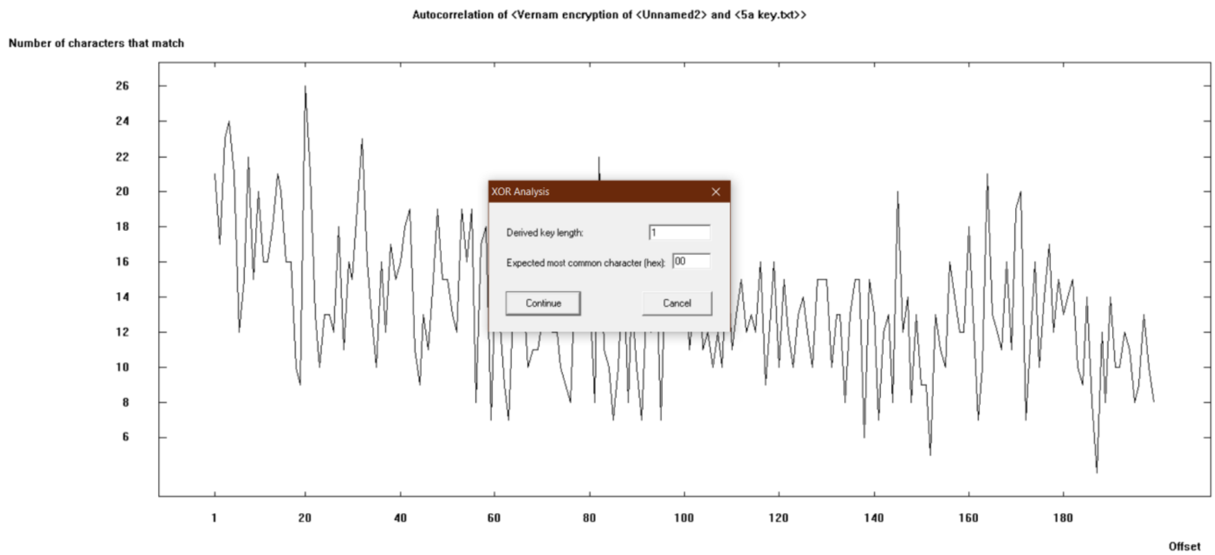
b. Use your full name as the OTP key and encrypt the same plaintext with the same algorithm. (don't forget to write the key in your answer file)



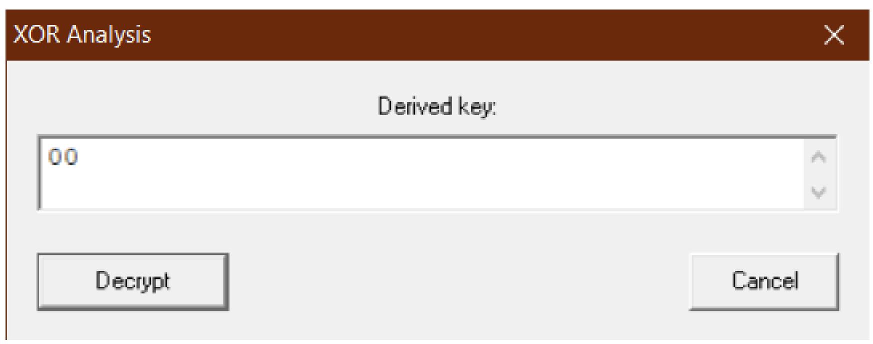
Key = 'MAHYARONSORI'

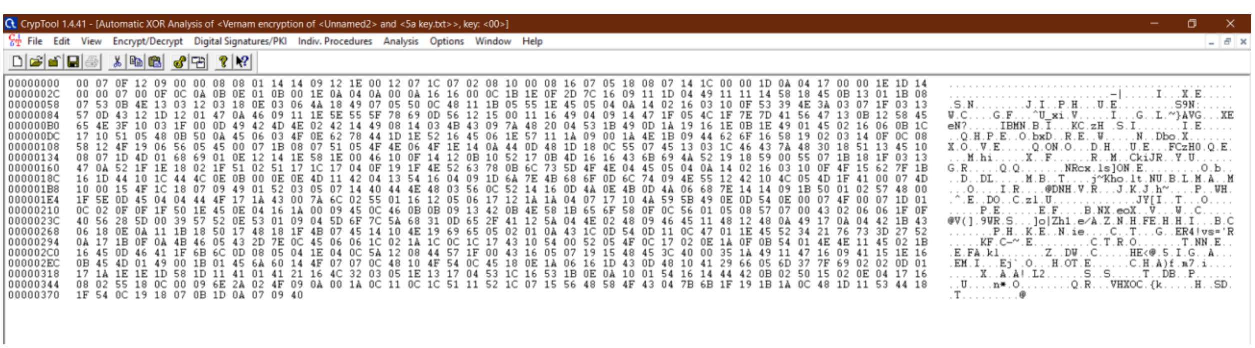


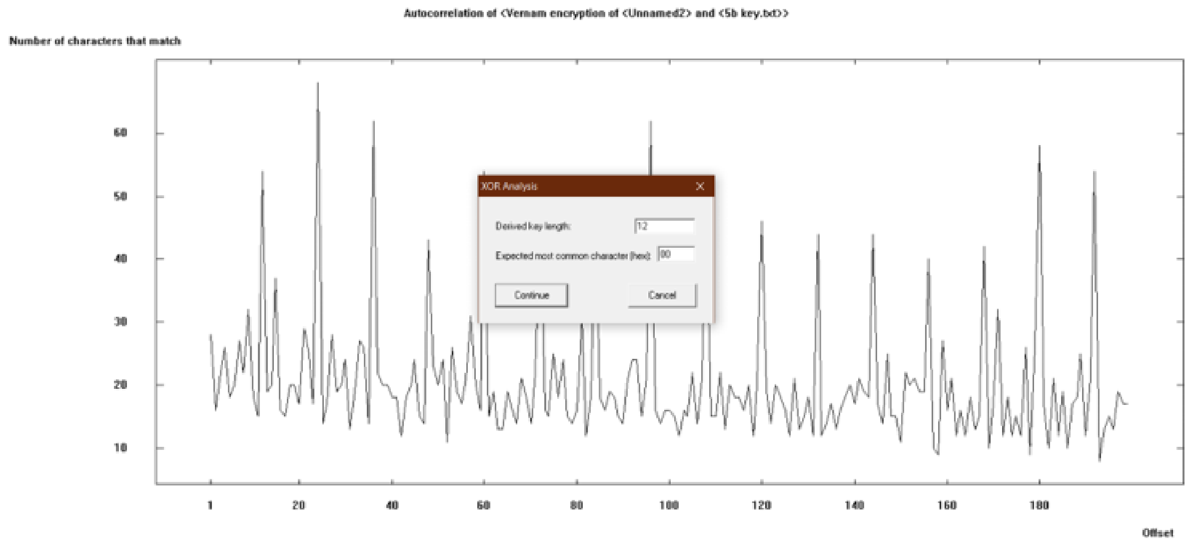
c. Try to find the OTP key of each of the two cipher texts in the previous parts using the Cryptool analytical tools and put the results, especially the predicted length of the key, in your answer file; Compare the security of the keys based on the results of analysis. (To do this part select Analysis > Symmetric Encryption > cipher text only > XOR)



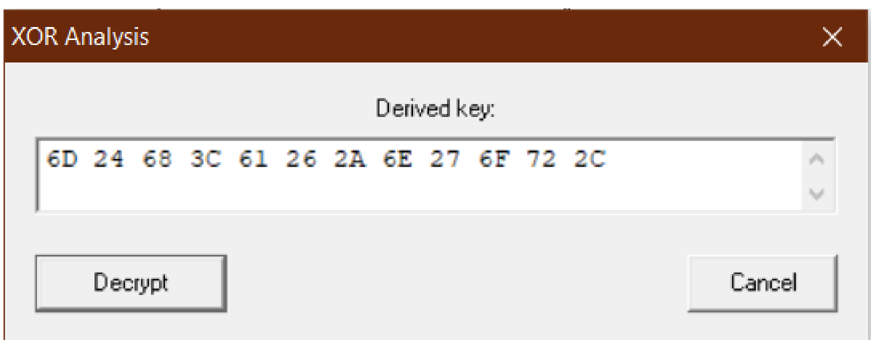
Predicted Key Length = 1

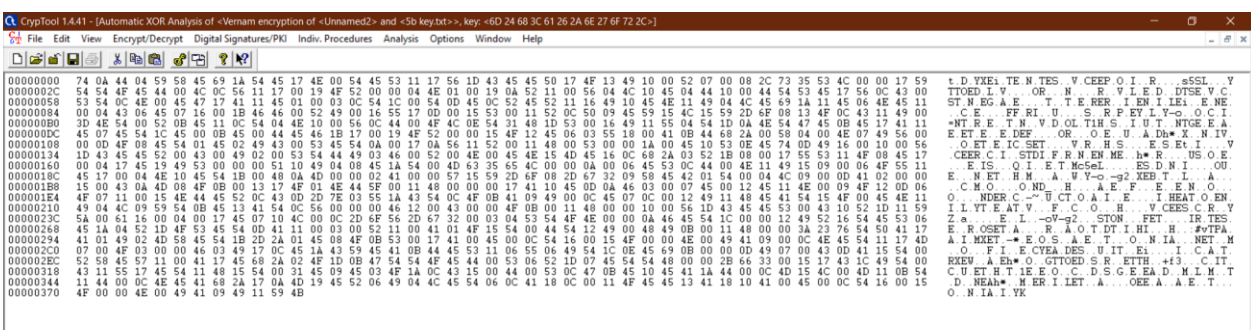






Predicted Key Length = 12





علت اینکه طول کلید را در حالت اول ۱ حدس زده است به این دلیل است که برنامه نتوانسته کلید را تشخیص دهد. در حالت دوم طول کلید را درست حدس میزند اما خود کلید را اشتباه تشخیص می دهد.